Problem 1c.
The results only slightly differ, the SE is centered on the wrong value, ~1.3 above the truth. It's also wider, as the omitted z is absorbed into the residual.
Statistical tests only assess sampling uncertainty, they cannot detect or fix bias from a misspecified model, and more data would not help either, as the SE will shrink while the bias stays.

Problem 2a.
I chose the following as my hypotheses: null hypothesis: $\mu_{manual} - \mu_{automatic} = 0$; Hypothesis: $\mu_{manual} - \mu_{automatic} \neq 0$.
I chose a t-test rather than a z-test, as the population SD is unknown and the sample is small. I used the Welch version because the group SDs differ, so equal variances should not be assumed.
$p < 0.05$, so we can reject the null hypothesis. Statistically, if transmission had no relationship with fuel economy, a gap this large would happen about 0.14% of the time. Substantively, manual cars in this sample get roughly 7 more mpg, roughly 42% more than automatic cars.

Problem 2c.
All three reject tests reject the null hypothesis at 0.05 and at 0.01. The permutation test and lm agree closely, so the pooled-variance assumptions did not distort things much. The Welch p offered more conservative results because it allows unequal group variances.